

TIM 172B, Lecture # 18, MOT-II (SCM)

3/5/26

Agenda

Two remaining topics

1. Facilities

2. Transportation

Coverage: First, Transportation; then Facilities

3. HW # 7 was posted, Thursday, last week
— Actively work on it.

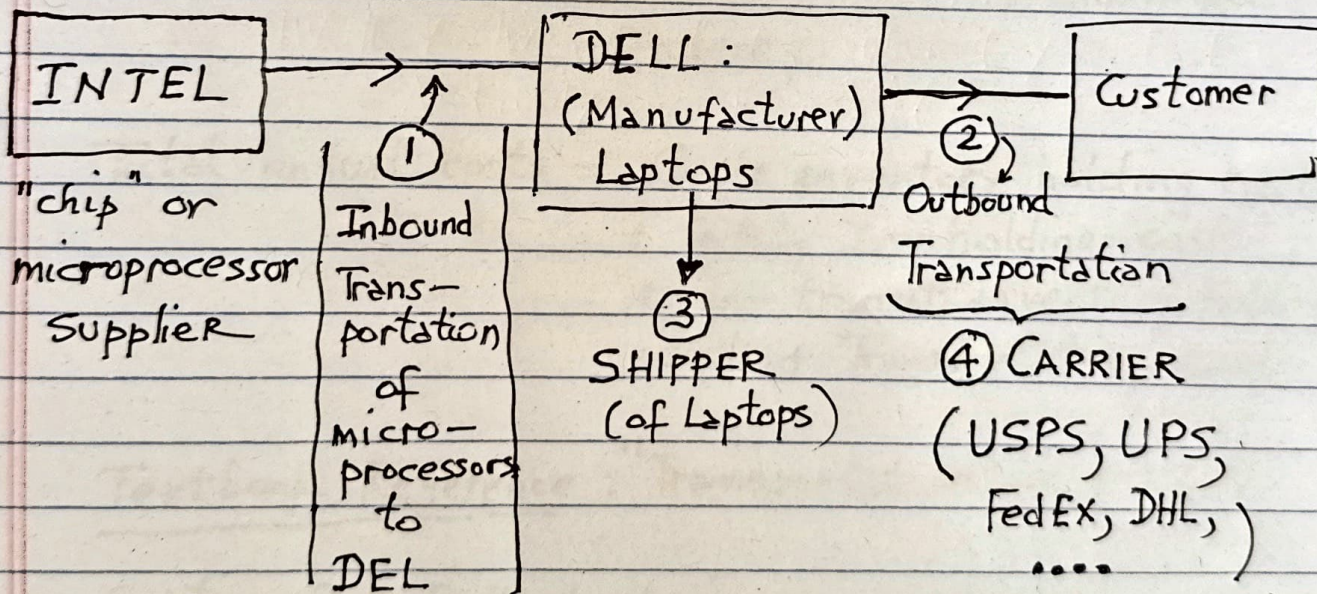
1. Transportation

Your action plan:

- Read the "Transportation" chapter in the text.
- Do the discussion problem: HW #7, Prob. 1
- Do HW #7, Prob. 7: "Books-on-Line" (Amazon)
- Read /study the "Transportation Lecture Notes" on Canvas

Notation (or Terminology)

Consider Dell's "Build-to-order" SC



Transportation Driver Design

5-step process described in more detail in The "Transportation Lecture Notes" on Canvas

For us, only the first 3-steps are important

Step 1 : Q : What modes of transportation should be used?

A : Use the mode of transportation for which the total annual cost is a minimum $\Rightarrow$ maximize efficiency

Total annual costs = Cycle inventory holding cost
+ Safety Inv. holding cost
+ In-transit inventory holding cost + Transportation cost

Textbook Reference : "Transportation in a SC"

Section 2 of the text is a useful discussion of the "pro & cons" of various modes of transportation : air, land (rail, trucks, ...), water.

For Step 1 create a table of options

- see Lecture Notes on Canvas

- study example 1 in the transportation chapter in the text

- set up & solve the problem in Excel (spread-sheet)

Step 2 : Q : should inventory be aggregated spatially?

A : Yes, because aggregation reduces cycle inventory costs [L#11,12] & safety inventory costs [L#19,20].

Create a table of options :

- Study example 2 in the "transportation chapter"

- See Lecture Notes

Step 3 : Q: Should inventory be aggregated in time (temporal aggregation)?

A: Yes, because temporal aggregation reduces total cost

Create a table of options

- See Lecture Notes
- Study Example 3 in the text.
- "Excel" it!

2. Facilities

Action Plan:

- Read "Facilities" chapter in the text ("Network Design in the SC")
- Do the discussion problems: HW #7, Prob 2
- Do the "SOLVER" tutorial on Canvas.
- Do the worked example in the text :
HW #7, Prob. 5
- Do HW #7, Prob. 6 ("Dry Ice"): set up, in-class, today.

Important considerations :

1. Facility Role or function : manufacturing ?
assembly ? warehouse ?
retail store ? ,
2. Location : where ? (Strategic)
3. Capacity : How big ?
4. Market : what markets or regions of demand should the facility serve.

A central issue in Facilities design is the Capacitated Plant Location Model (CPLM) :

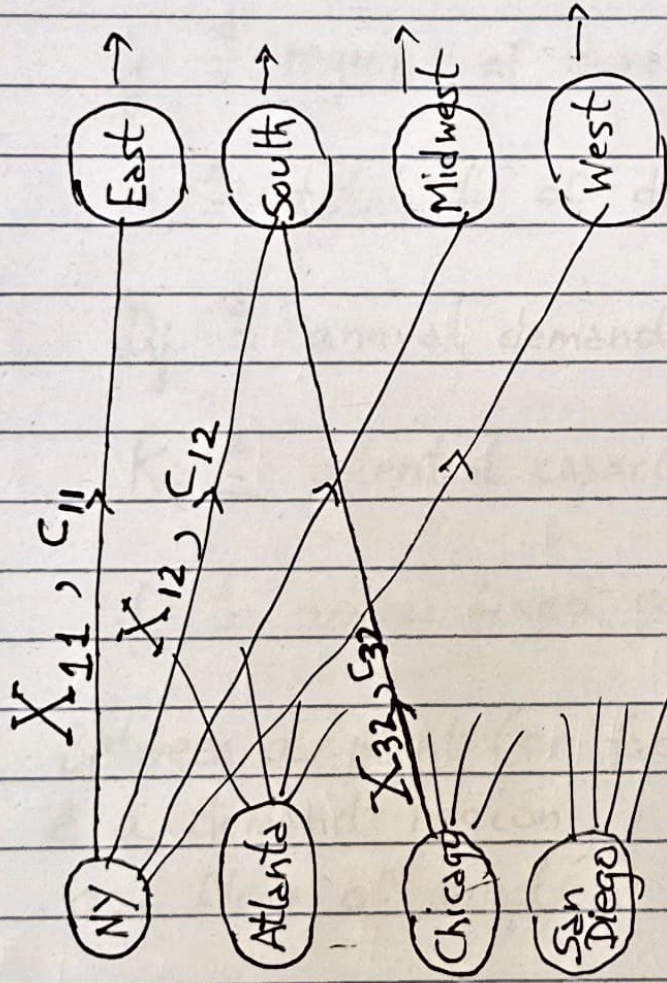
Determine the location & capacity of the facilities (aka plants) to minimize total cost $\Rightarrow$ maximize efficiency.

DRY ICE (HW#7, Prob. 6)

PLANTS

i	y_i	k_i	f_i
1	y_1	k_1	f_1
2	y_2	k_2	f_2
3	y_3	k_3	f_3
4	y_4	k_4	f_4

$n = \text{total \# of plants}$
 $= 4$



Markets or Regions of Demand

j	D_j
1	D_1
2	D_2
3	D_3
4	D_4

$m = \text{total number of markets or regions of demand}$
 $= 4$

$i \triangleq$ plant, i , ($i=1, 2, \dots, n$)

$n \triangleq$ total # of plants (or facilities)

$j \triangleq$ regions of demand, or markets, ($j=1, 2, \dots, m$)

$m \triangleq$ total # of demand regions

$D_j \triangleq$ annual demand for region j , ($j=1, 2, \dots, m$)

$K_i \triangleq$ potential capacity of plant i , ($i=1, 2, \dots, n$)

$f_i \triangleq$ annual fixed cost of operating plant i , ($i=1, 2, \dots, n$)

Between a plant (or facility), i , ($i=1, 2, \dots, n$) & a demand region, j , ($j=1, 2, \dots, m$) we have a flow of supply

$X_{ij} \triangleq$ quantity of items transported from plant i to demand region, j

$c_{ij} \triangleq$ total cost per items
material cost + manufacturing cost
+ inventory holding cost + transportation cost +

$Y_i \triangleq$ plant location decision variable

$$Y_i : \begin{cases} = 1 & \text{locate plant at location } i \\ = 0 & \text{no plant at location } i \end{cases}$$

Informal problem statement:

Determine the plant locations, Y_i , capacity K_i ,
& the supply flows X_{ij} , to minimize the
total annual cost, C

Process (or Plan or Approach or Procedure):

1. Determine the total cost, C

$$C = \sum_{i=1}^n Y_i f_i + \sum_{i=1}^n \sum_{j=1}^m X_{ij} c_{ij} \rightarrow (1)$$

Goal: minimize, C , given by (1).